

DE analysis

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```
# load the packages
library(ggplot2); theme_set(theme_bw())
library(cluster)
library(salvo)
library(ggpubr)
library(dplyr)
library(fdrtool)
library(Seurat)

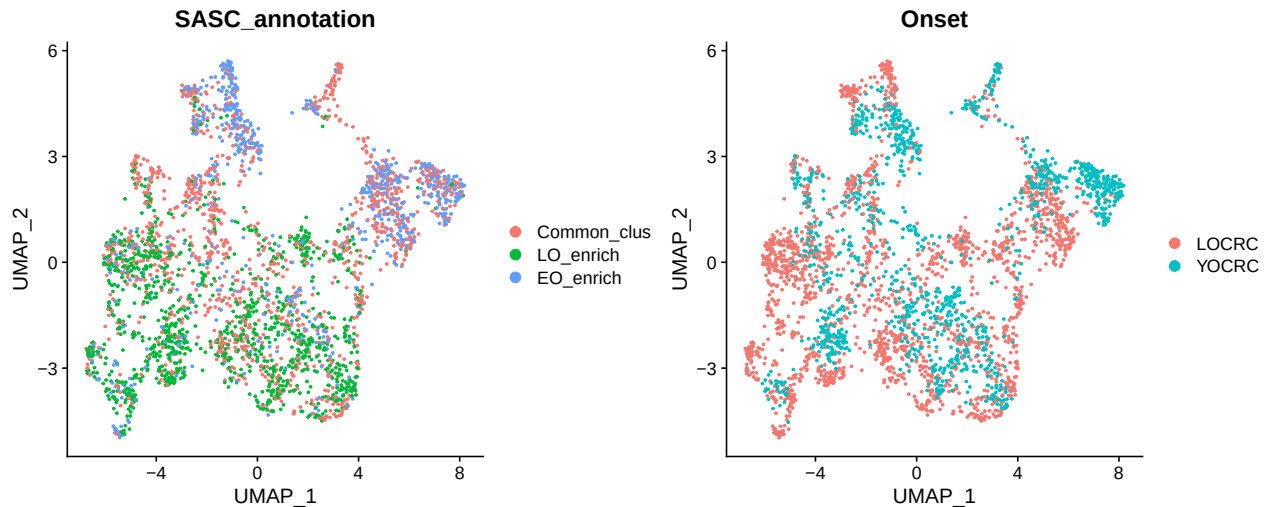
# read the model output
if (file.exists("../output/realdata/data/SASC_output.rds")) {
  output = readRDS(file = "../output/realdata/data/SASC_output.rds")
  clusterinfo = readRDS(file = "../output/realdata/data/clusterinfo.rds")
} else {
  cat("***Please first run 'SASC_CRC.Rmd' for model-based inference.***")
}
```

```
# print the clustering results
output$SASC_cluster = factor(output$SASC_cluster, c(1,6,2:5,7,8,10,9,11))
table(output$SASC_cluster, output$annotation_final)
```

```
##
##      cd4T_hp cd4T_other cd4T_rg CD8T_em CD8T_ex CD8T_other
##  1         8         12      160         0         0         0
##  6        122         22      279         0         0         0
##  2        208         48       30         0         0         0
##  3         95         13       91         0         0         0
##  4        405         13        6         0         0         0
##  5         20        90        4         0         0         0
##  7          0          0         0       273        72        28
##  8          0          0         0        67        56        54
## 10          0          0         0        26       189         3
##  9          0          0         0       271        12       339
## 11          0          0         0        20        33        70
```

```
# Assign annotation results
SASC_annotation = rep("Common_clus", dim(output)[2])
SASC_annotation[which(output$SASC_cluster == 4 | output$SASC_cluster == 5 | output$SASC_cluster == 9)] = "LO_enrich"
SASC_annotation[which(output$SASC_cluster == 6 | output$SASC_cluster == 10 | output$SASC_cluster == 11)] = "EO_enrich"
output$SASC_annotation = factor(SASC_annotation, levels = c("Common_clus", "LO_enrich", "EO_enrich"))

# Visualize the results
DimPlot(output, group.by = "SASC_annotation")+
  DimPlot(output, group.by = "Onset")
```



```

Idents(output) = output$T_cell_type
CD4 = subset(output, idents = "CD4")
CD8 = subset(output, idents = "CD8")

```

The CD4 T helper cells

```

# Identify conditional specific genes for CD4 subtypes
Idents(CD4) = CD4$annotation_final
CD4_help = subset(CD4, idents = "cd4T_hp")
Idents(CD4_help) = CD4_help$SASC_annotation
output.markers <- FindAllMarkers(CD4_help, only.pos = TRUE, min.pct = 0.25, logfc.threshold = 0.25)

```

```
## Calculating cluster Common_clus
```

```
## Calculating cluster LO_enrich
```

```
## Calculating cluster EO_enrich
```

```

top2 = output.markers %>%
  group_by(cluster) %>%
  slice_max(n = 5, order_by = avg_log2FC)
top2

```

```
## # A tibble: 15 x 7
```

```
## # Groups:   cluster [3]
```

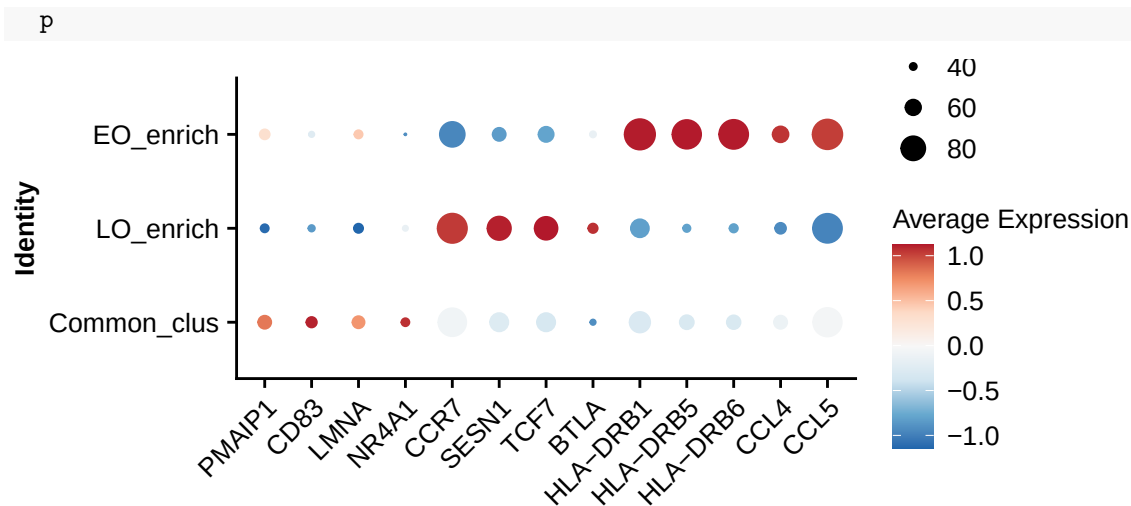
	p_val	avg_log2FC	pct.1	pct.2	p_val_adj	cluster	gene
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<chr>
## 1	4.59e- 7	0.882	0.54	0.433	5.75e- 3	Common_clus	PMAIP1
## 2	3.00e- 4	0.813	0.479	0.38	1 e+ 0	Common_clus	CD83
## 3	1.19e- 3	0.749	0.518	0.444	1 e+ 0	Common_clus	LMNA
## 4	1.40e- 3	0.647	0.421	0.338	1 e+ 0	Common_clus	NR4A1
## 5	8.23e- 3	0.542	0.907	0.921	1 e+ 0	Common_clus	CCL5
## 6	9.74e- 8	0.662	0.925	0.871	1.22e- 3	LO_enrich	CCR7
## 7	7.12e- 3	0.559	0.4	0.335	1 e+ 0	LO_enrich	ARHGAP5
## 8	4.06e- 6	0.526	0.786	0.628	5.08e- 2	LO_enrich	SESN1
## 9	1.49e- 8	0.510	0.772	0.642	1.87e- 4	LO_enrich	TCF7
## 10	1.11e- 3	0.506	0.454	0.365	1 e+ 0	LO_enrich	BTLA
## 11	6.65e-39	2.04	0.951	0.682	8.33e-35	EO_enrich	HLA-DRB1

```
## 12 2.04e-14      1.78  0.738 0.51   2.55e-10 EO_enrich  HLA-DRA
## 13 4.90e-42      1.61  0.91  0.473  6.14e-38 EO_enrich  HLA-DRB5
## 14 3.07e-42      1.55  0.918 0.484  3.85e-38 EO_enrich  HLA-DRB6
## 15 6.34e- 4      1.52  0.607 0.512  1    e+ 0 EO_enrich  CCL4
```

```
# Print our the final DE genes that we are interested in.
```

```
final_list=c("PMAIP1","CD83","LMNA","NR4A1","CCR7","SESN1","TCF7","BTLA","HLA-DRB1","HLA-DRB5","HLA-DRB6")
p =
DotPlot(CD4_help, features = unique(final_list), dot.scale=5, cols="RdBu") +
  theme(title=element_text(size=10), axis.text.x=element_text(size=10, angle=45, hjust=1),
        axis.title.x=element_text(size=0), axis.text.y=element_text(size=10),
        axis.title.y=element_text(size=10, face="bold"), legend.position = "right",
        legend.text=element_text(size=10), legend.title=element_text(size=10))
```

```
## Warning: Scaling data with a low number of groups may produce misleading
## results
```



The CD4 Treg cells

```
# Subset the Tregs
CD4_reg = subset(CD4, idents = "cd4T_rg")
Idents(CD4_reg) = CD4_reg$SASC_annotation
output.markers <- FindAllMarkers(CD4_reg, only.pos = TRUE, min.pct = 0.25, logfc.threshold = 0.25)
```

```
## Calculating cluster Common_clus
```

```
## Calculating cluster LO_enrich
```

```
## Calculating cluster EO_enrich
```

```
top2 = output.markers %>%
  group_by(cluster) %>%
  slice_max(n = 4, order_by = avg_log2FC)
top2
```

```
## # A tibble: 12 x 7
```

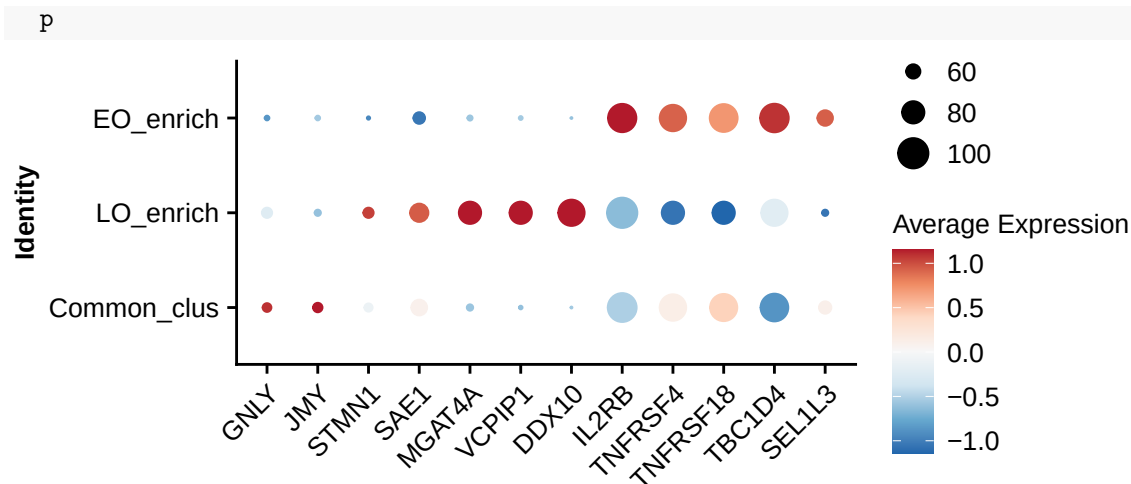
```
## # Groups:   cluster [3]
```

```
##       p_val avg_log2FC pct.1 pct.2 p_val_adj cluster  gene
##       <dbl>    <dbl> <dbl> <dbl>    <dbl> <fct>    <chr>
## 1 0.000100      1.27  0.448 0.332      1 Common_clus STMN1
```

```
## 2 0.00854      1.13 0.463 0.367      1      Common_clus GNLY
## 3 0.000614     0.534 0.48 0.363      1      Common_clus JMY
## 4 0.00416     0.521 0.377 0.284      1      Common_clus NUSAP1
## 5 0.00197     4.26 0.4 0.107      1      LO_enrich IL17A
## 6 0.000211     1.65 0.8 0.388      1      LO_enrich MGAT4A
## 7 0.000231     1.58 0.8 0.338      1      LO_enrich VCPIP1
## 8 0.00000465    1.43 0.9 0.291    0.0583 LO_enrich DDX10
## 9 0.00282     0.543 0.961 0.942      1      EO_enrich LTB
## 10 0.0000482    0.444 0.957 0.935    0.603 EO_enrich TBC1D4
## 11 0.00501     0.359 0.631 0.546      1      EO_enrich SEL1L3
## 12 0.000182     0.347 0.95 0.966      1      EO_enrich IL2RB
```

```
# Print our the final DE genes that we are interested in.
final_list=c("GNLY","JMY","STMN1","SAE1","MGAT4A","VCPIP1","DDX10","IL2RB","TNFRSF4","TNFRSF18","TBC1D4","SEL1L3")
p =
DotPlot(CD4_reg, features = final_list, dot.scale=5, cols="RdBu") +
  theme(title=element_text(size=10), axis.text.x=element_text(size=10, angle=45, hjust=1),
        axis.title.x=element_text(size=0), axis.text.y=element_text(size=10),
        axis.title.y=element_text(size=10, face="bold"), legend.position = "right",
        legend.text=element_text(size=10), legend.title=element_text(size=10))
```

```
## Warning: Scaling data with a low number of groups may produce misleading
## results
```



The CD8 T effector memory cells

```
# subset cell type
Idents(CD8) = CD8$annotation_final
CD8_em = subset(CD8, idents = "CD8T_em")
Idents(CD8_em) = CD8_em$SASC_annotation
output.markers <- FindAllMarkers(CD8_em, only.pos = TRUE, min.pct = 0.25, logfc.threshold = 0.25)
```

```
## Calculating cluster Common_clus
```

```
## Calculating cluster LO_enrich
```

```
## Calculating cluster EO_enrich
```

```
top2 = output.markers %>%
  group_by(cluster) %>%
```

```

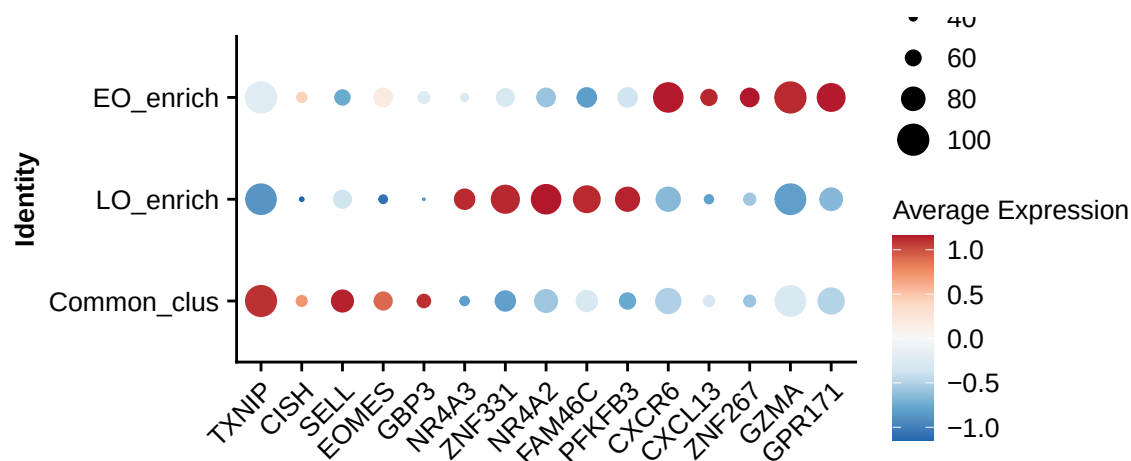
slice_max(n = 5, order_by = avg_log2FC)
top2

## # A tibble: 15 x 7
## # Groups:   cluster [3]
##       p_val avg_log2FC pct.1 pct.2 p_val_adj cluster  gene
##       <dbl>    <dbl> <dbl> <dbl>    <dbl> <fct>    <chr>
## 1 2.21e- 7      0.923 0.997 0.987 2.76e- 3 Common_clus TXNIP
## 2 1.08e- 4      0.856 0.474 0.328 1 e+ 0 Common_clus CISH
## 3 9.83e- 5      0.717 0.762 0.647 1 e+ 0 Common_clus SELL
## 4 2.46e- 9      0.704 0.659 0.451 3.08e- 5 Common_clus EOMES
## 5 2.18e-11      0.641 0.538 0.29 2.72e- 7 Common_clus GBP3
## 6 7.99e-17      1.01 0.72 0.427 1.00e-12 LO_enrich NR4A3
## 7 4.66e-18      0.965 0.919 0.71 5.84e-14 LO_enrich ZNF331
## 8 2.25e-18      0.932 0.956 0.772 2.82e-14 LO_enrich NR4A2
## 9 6.76e-12      0.900 0.886 0.738 8.47e- 8 LO_enrich FAM46C
##10 1.06e-11      0.872 0.815 0.622 1.33e- 7 LO_enrich PFKFB3
##11 1.54e- 5      1.05 0.957 0.833 1.92e- 1 EO_enrich CXCR6
##12 5.17e- 3      0.994 0.609 0.458 1 e+ 0 EO_enrich CXCL13
##13 8.79e- 4      0.982 0.674 0.501 1 e+ 0 EO_enrich ZNF267
##14 1.26e- 5      0.977 1 0.989 1.58e- 1 EO_enrich GZMA
##15 4.09e- 4      0.891 0.913 0.83 1 e+ 0 EO_enrich GPR171

# Print our the final DE genes that we are interested in.
p =
  DotPlot(CD8_em, features = unique(top2$gene), dot.scale=5, cols="RdBu") +
  theme(title=element_text(size=10), axis.text.x=element_text(size=10, angle=45, hjust=1),
        axis.title.x=element_text(size=0), axis.text.y=element_text(size=10),
        axis.title.y=element_text(size=10, face="bold"), legend.position = "right",
        legend.text=element_text(size=10), legend.title=element_text(size=10))

## Warning: Scaling data with a low number of groups may produce misleading
## results
p

```



The CD8 T exhausted cells

```

# subset cell type
CD8_ex = subset(CD8, identfs = "CD8T_ex")
Identfs(CD8_ex) = CD8_ex$SASC_annotation
output.markers <- FindAllMarkers(CD8_ex, only.pos = TRUE, min.pct = 0.25, logfc.threshold = 0.25)

## Calculating cluster Common_clus
## Calculating cluster LO_enrich
## Calculating cluster EO_enrich

top2 = output.markers %>%
  group_by(cluster) %>%
  slice_max(n = 5, order_by = avg_log2FC)
top2

## # A tibble: 15 x 7
## # Groups:   cluster [3]
##       p_val avg_log2FC pct.1 pct.2 p_val_adj cluster      gene
##       <dbl>      <dbl> <dbl> <dbl>    <dbl> <fct>      <chr>
## 1 0.0000794      2.24 0.359 0.209 0.995   Common_clus RRM2
## 2 0.000000129    1.91 0.969 0.846 0.00162   Common_clus TUBA1B
## 3 0.000249      1.69 0.734 0.645 1       Common_clus STMN1
## 4 0.000758      1.20 0.5   0.368 1       Common_clus TYMS
## 5 0.0000511     1.17 0.391 0.231 0.640   Common_clus TOP2A
## 6 0.000353      2.18 0.833 0.517 1       LO_enrich   KLRC1
## 7 0.000298      2.05 0.917 0.689 1       LO_enrich   FOS
## 8 0.0000430     1.62 1      0.589 0.538   LO_enrich   ZNF331
## 9 0.00000721    1.55 1      0.974 0.0903   LO_enrich   TNFAIP3
## 10 0.0000140    1.53 1      0.623 0.175   LO_enrich   BTG2
## 11 0.00000000743 1.14 0.545 0.264 0.0000931 EO_enrich   FCRL3
## 12 0.0000299    0.979 0.757 0.557 0.374   EO_enrich   KLRG1
## 13 0.000164     0.817 0.761 0.65 1       EO_enrich   TNFSF4
## 14 0.000000669 0.784 0.45 0.214 0.00838   EO_enrich   INPP1
## 15 0.0000218    0.718 0.919 0.85 0.273   EO_enrich   TTN

p =
  DotPlot(CD8_ex, features = unique(top2$gene), dot.scale=5, cols="RdBu") +
    theme(title=element_text(size=10), axis.text.x=element_text(size=10, angle=45, hjust=1),
          axis.title.x=element_text(size=0), axis.text.y=element_text(size=10),
          axis.title.y=element_text(size=10, face="bold"), legend.position = "right",
          legend.text=element_text(size=10), legend.title=element_text(size=10))

## Warning: Scaling data with a low number of groups may produce misleading
## results

p

```

